ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CCS6611 COMMUNICATION NETWORKS

FINAL EXAMINATION

WEDNESDAY 4TH DECEMBER 2024

14:00 HRS TO 17:00 HRS

TIME ALLOWED : (3 HRS WRITING – 5 MIN READING)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

QUESTION 1

ZCAS University, a beacon of academic excellence, is embarking on an ambitious expansion. A new campus, nestled in a remote region, promises to open doors for aspiring students and propel the institution to new heights. However, this expansion presents a unique challenge: bridging the digital divide. With limited network connectivity in the remote location, ensuring seamless communication, access to resources, and a robust digital learning environment is paramount.

As a senior network engineer, you are at the forefront of this technological endeavor. Your task is to design a resilient and secure network infrastructure that can withstand the test of time and distance. To this effect, the management engaged an independent consultant to carry out a feasibility study and provide a summary report highlighting activities that would lead to a successful implementation of a reliable solution. After a successful situation analysis and study, you have been provided with a summary report highlighting the following recommendations:

1. **Architecting a Robust Network:**
   - Design a network topology that balances cost-effectiveness and performance.
   - Consider hybrid solutions combining wired and wireless technologies to optimize connectivity.
   - Implement redundancy and failover mechanisms to ensure uninterrupted service.

2. **Conquering the Connectivity Challenge:**
   - Explore satellite internet, fiber-optic, or wireless broadband options, evaluating their reliability and cost-effectiveness.
   - Implement effective network management tools to monitor and troubleshoot network issues remotely.

3. **Fortifying the Digital Fortress:**
   - Employ robust security measures, including firewalls, intrusion detection systems, and encryption protocols.
   - Implement strong access controls to safeguard sensitive data and intellectual property.
   - Stay updated on the latest cybersecurity threats and vulnerabilities.

4. **Enabling Remote Collaboration:**

- Design a secure remote access solution for faculty and staff to connect to the main campus network.
- Implement collaboration tools to facilitate seamless communication and knowledge sharing.

5. **Elevating the Learning Experience:**
   - Ensure high-speed internet access for students and faculty.
   - Implement virtual learning platforms and online resources to enhance the learning experience.
   - Provide technical support to students and faculty to maximize the benefits of technology.

Based on the recommendations, the management is very confident and highly expectant that a successfully executed network expansion, will not only empower ZCAS University to reach new heights but also contribute to the digital transformation of education in the region. A summary report by the consultant highlights the following as your main activities.

i. **Network Topology:** Designing a suitable network topology for the branch, considering factors such as scalability, reliability, and cost-effectiveness.

ii. **Connectivity:** Proposed a solution for providing reliable internet connectivity to the remote branch, considering options like satellite, fiber, or wireless technologies.

iii. **Security:** Implementation of a robust security measures to protect the network and sensitive data, including firewalls, intrusion detection systems, and encryption protocols.

iv. **Remote Access:** Designing a secure remote access solution for employees to connect to the main office network, ensuring data confidentiality and integrity.

v. **Network Monitoring:** Implementing a network monitoring tools to track performance and troubleshoot issues proactively.

Additional Considerations:

- **Scalability:** Ensure the network can accommodate future growth and expansion.
- **User Experience:** Optimize the network for optimal user experience, including fast response times and minimal downtime.

Questions

i. Create a detailed network diagram illustrating the proposed network topology and device configurations. (10 Marks)

ii. Provide detailed specifications for hardware components (e.g., routers, switches, firewalls) and software configurations (e.g., routing protocols, security policies). (10 Marks)

iii. Outline a step-by-step implementation plan, including tasks, timelines, and resource allocation. (10 Marks)

iv. Detail the specific security measures to be implemented, including firewall rules, intrusion detection system configurations, and encryption protocols. (10 Marks)

v. Outline a plan for ongoing network monitoring, troubleshooting, and maintenance. (10 Marks)

(Total: 50 marks)

SECTION B: ATTEMPT ANY TWO (2) QUESTIONS IN THIS SECTION

QUESTION 2

a. Using suitable examples, explain the difference between a virtual circuit and a datagram? (10 Marks)

b. Describe common network security threats and discuss the importance of implementing security measures like firewalls, intrusion detection systems, and encryption. (15 Marks)

(25 marks)

QUESTION 3

a. Illustrate and explain the layered architecture of the TCP/IP model and state the role of each layer and how they interact. (15 Marks)

b. Discuss the concept of congestion control in TCP and the mechanisms used to manage it. (10 Marks)

(25 marks)

QUESTION 4

Discuss the factors that affect network performance and techniques to improve the performance. (15 Marks)

Describe the common tools and techniques used for network troubleshooting. (10 Marks)

(25 marks)

END OF EXAMINATION